



VVCE

Vidyavardhaka College Of Engineering, Mysuru

Affiliated by VTU and approved by AICTE Accredited by NAAC & NBA with 'A' grade

Gokulam 3rd stage, Mysuru , Karnataka.



Department of CSE (Artificial Intelligence and Machine Learning)

**Course Name : AWS Academy AI Foundations
(BCLAI656)**

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INTRODUCTION

About the Course:

- Introduces the core concepts of Artificial Intelligence and Generative AI.
- Covers Machine Learning, Foundation Models, and Prompt Engineering.
- Explains Responsible AI principles and security considerations.
- Provides exposure to AWS services used for AI development.
- Connects AI fundamentals with real-world cloud-based applications.

Importance of Course:

- Generative AI is transforming industries and business processes.
- Organizations are increasingly adopting AI-powered solutions.
- AI knowledge is becoming an essential skill for technology professionals.
- AWS provides scalable and secure platforms for AI innovation.
- Helps learners understand modern AI technologies and their applications.



OBJECTIVES OF THE COURSE

Core Objectives:

AI and ML Objectives

Objective

What It Means



Understand AI and ML concepts

Grasp the basics of how machines learn



Explore Generative AI and FMs

Learn how AI creates new content



Learn Prompt Engineering

Master talking to AI effectively



Understand Responsible AI

Build AI that is fair and safe



Study Security & Compliance

Protect AI systems and data



Gain AWS AI exposure

Work with AWS AI services



Prepare for Certification

Get ready for AWS AI exams



Sub-Course 1 - Welcome to AWS Academy Generative AI Foundation

Topics Covered :

AWS Academy Program

- Introduces industry-aligned cloud and AI learning designed to prepare students for real-world technology careers.

Course Overview

- Provides a roadmap of the Generative AI Foundations course, including AI concepts, Foundation Models, Prompt Engineering, and AWS AI services.

Generative AI on AWS

- Explores how AWS supports the development and deployment of Generative AI applications using cloud-based tools and services.

Learning Resources and Support

- Familiarizes students with AWS Academy resources, student guides, assessments, and hands-on learning opportunities.

Skills Gained :

- Understanding the purpose and structure of AWS Academy programs.
- Familiarity with the Generative AI Foundations learning pathway.
- Awareness of AWS tools and services used in AI development.
- Ability to navigate AWS learning resources and course materials.
- Understanding how cloud technologies support modern AI applications.



AWS Academy helps close the gap

AWS Academy provides higher education institutions with free, ready-to-teach cloud computing curriculum that prepares students to pursue industry-recognized certifications and careers in the cloud.



Curriculum
developed and
maintained by AWS



Post-secondary
program integration



Aligned to
AWS Certification



Employment
readiness



Sub-Course 2 – Introducing AI

Topics Covered :

What is Artificial Intelligence?

-> AI is the science of creating machines that simulate human intelligence. It's about making computers think, learn, and make decisions like people do.

Machine Learning (ML)

-> ML is a subset of AI where algorithms learn from data and experience. Instead of being told what to do, ML systems find patterns on their own.

Deep Learning (DL)

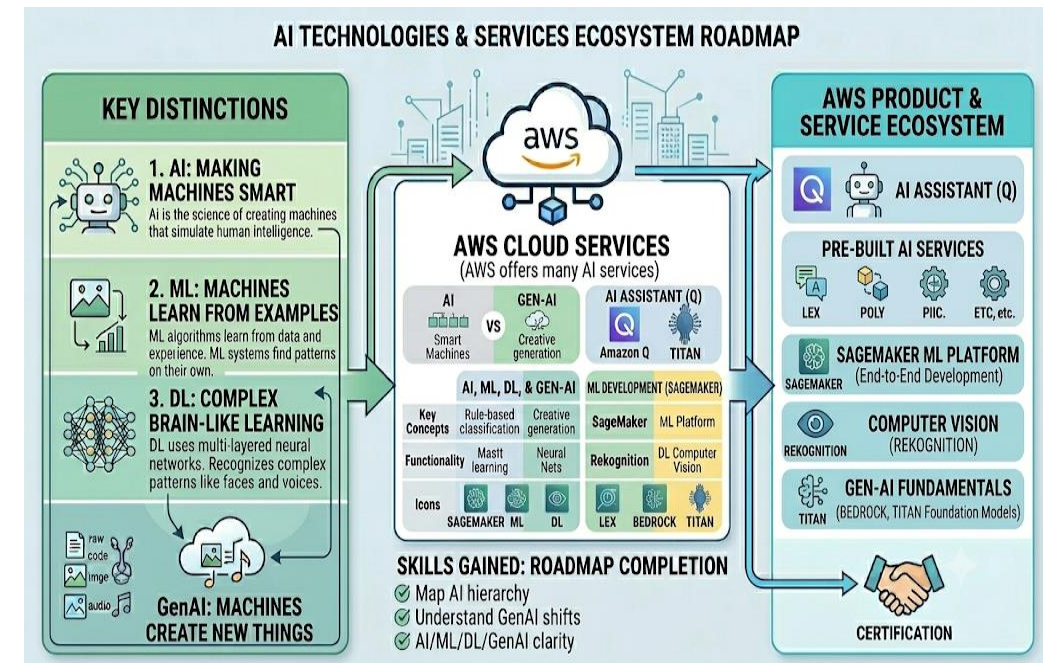
-> DL uses multi-layered neural networks inspired by the human brain. These networks can recognize complex patterns like faces, voices, and text.

Generative AI

-> GenAI is a subset of DL that creates entirely new content - text, images, music, code - rather than just classifying or predicting.

Skills Gained :

- Ability to map the evolution and hierarchy of AI technologies.
- Understanding of the fundamental shifts from traditional AI to Modern Generative models.
- Clarity on tech distinctions between AI , ML , DL and Gen-AI.



Sub-Course 3 – Introducing Generative AI

Topics Covered :

What are Foundation Models (FMs)?

-> Foundation Models are large-scale AI models pre-trained on massive amounts of data. They can perform many different tasks without being specially trained for each one.

Amazon Bedrock :

-> A managed service that gives you secure access to top Foundation Models through a simple API. It makes building generative AI applications easy and secure.

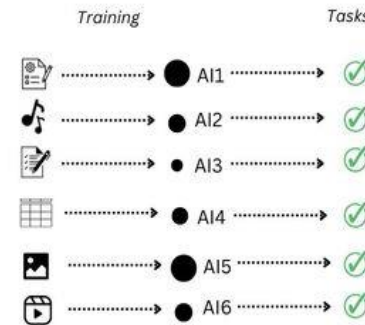
PartyRock :

-> A no-code playground where you can build AI apps in minutes using Amazon Bedrock. Great for experimenting and testing ideas quickly.

Skills Gained :

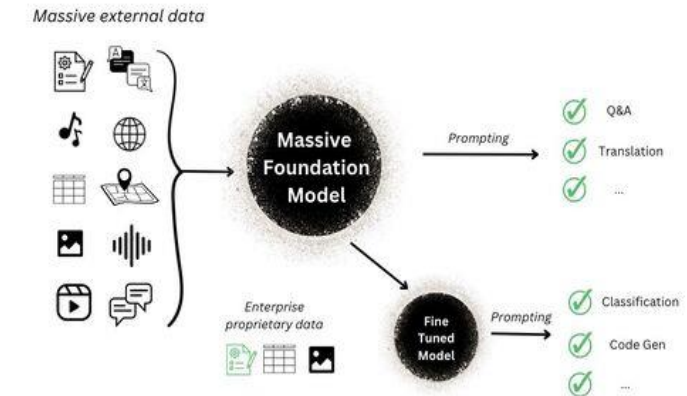
- Deep understanding of the Architecture and utility of Foundational model.
- Proficiency in using Amazon Bedrock for scalable AI application development.
- Practical skills in rapid prototyping with no code Ai tools

Traditional ML



- Individual siloed models
- Require task-specific training
- Lots of human supervised training

Foundation Models



- Massive multi-tasking model
- Adaptable with little or no training
- Pre-trained unsupervised learning



Sub-Course 4 – Using Prompts and Prompt Engineering

Topics Covered :

What is Prompt Engineering?

-> Prompt engineering is the art of crafting inputs that guide AI models to give you the outputs you want. It's like learning to speak the language of AI effectively.

Zero-Shot

Direct instruction without examples. AI rely solely on its pre-trained data.

Few-Shot

Providing a small set of input output examples to teach the model a specific pattern or style

Chain-of-Thought

Encouraging the model to articulate its reasoning process step by step, leading to more accurate and complex problem solving .

Skills Gained :

- Mastering in designing clear, effective prompts for driven AI tasks.
- Proficiency in applying advanced prompting techniques for optimal results.
- Ability to debug and refine prompts to achieve precise Ai responses.

Prompt Element Comparison

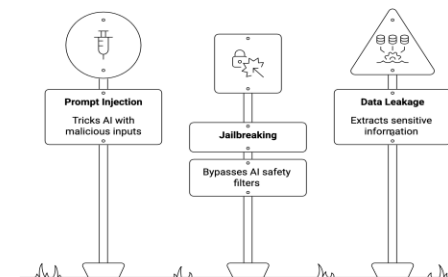
Element	What It Does
 Instruction	Tells the AI what to do
 Context	Gives background information
 Examples	Shows sample inputs and outputs
 Constraints	Sets limits on the response

Prompting Techniques

	 How It Works	 Best For
Zero-shot	Just ask, no examples	Simple tasks
Few-shot	Give 2-5 examples first	Specific formats
Chain-of-Thought	Ask for step-by-step reasoning	Complex problems

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AI Prompt Risks



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Sub-Course 5 – Working with Foundation Models

Topics Covered :

Model Selection

-> Choosing FM's based on Cost, Speed, Accuracy, Use cases.

Understanding Inference Parameters

Temperature

- Controls creativity and randomness.

Top-K

- Limits the number of possible next-word choices.

Top-P

- Selects words from the most probable choices.

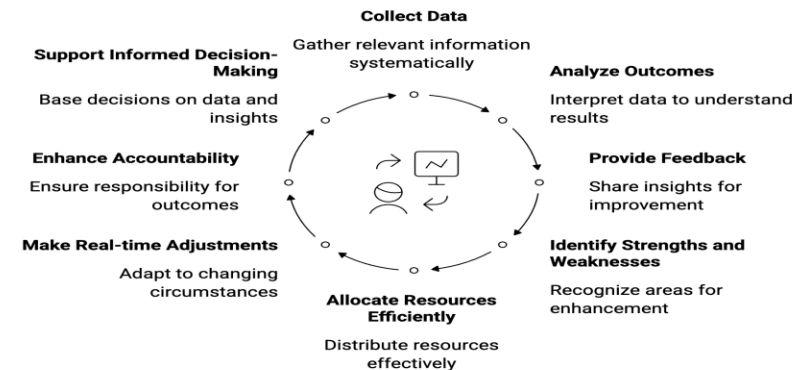
Why RAG Is Important ?

-> Retrieval Augmented Generation allows AI systems to retrieve information from trusted knowledge sources before generating responses. This improves factual accuracy and reduces hallucinations.

Skills Gained :

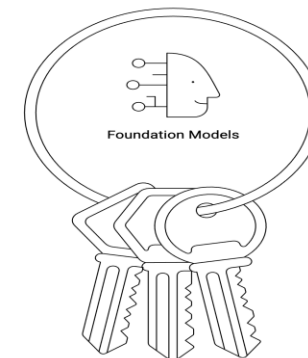
- Different Foundation Models are suitable for different use cases.
- RAG improves reliability without retraining models.
- Evaluation and monitoring are essential for maintaining model quality.

Evaluation and Monitoring Cycle



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Enhancing Foundation Models



- RAG**
Augments prompts with retrieved information to reduce false information and provide up-to-date answers.
- Fine-tuning**
Adapts models on specific data to tailor their responses to particular needs.
- Instruction Pre-training**
Trains models on specific instruction types to improve their understanding and execution of tasks.

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Sub-Course 6 – Practicing Generative AI Responsibly

Topics Covered :

Defining Responsible AI

-> Responsible AI means building AI systems that are ethical, fair, and transparent.

It's about making sure AI helps people without causing harm.

Fairness: Treats everyone equally and even

Transparency and explainability: know how AI make decisions, be able to explain well.

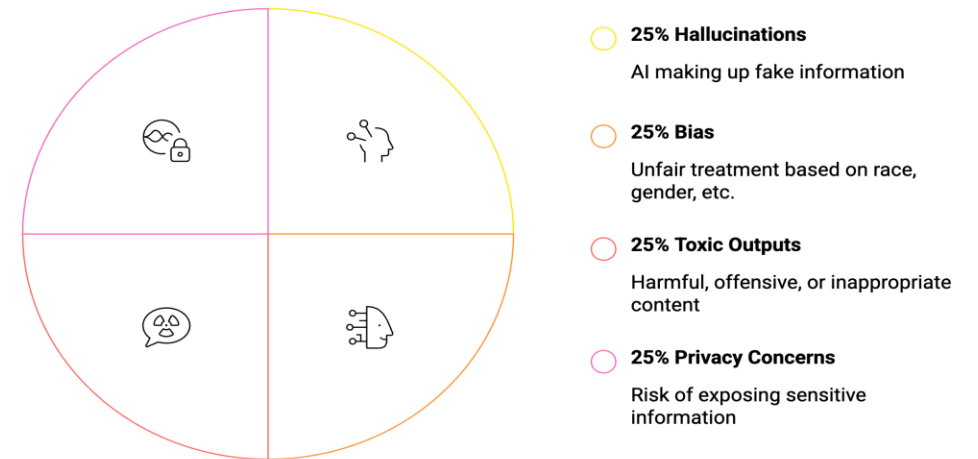
Privacy and Security: Protects people's data, Keep AI Systems safe.

Risks: Addressing potential issues like hallucinations, inherent biases from training data and toxic outputs , AWS Bedrock Guardrails help in preventing harmful content.

Skills Gained :

- Ability identify and mitigate ethical risks in AI development.
- Understanding of best practices for implementing responsible AI.
- Proficiency in utilizing AWS Guardrails for content moderation and safety.

Distribution of Generative AI Risks



Sub-Course 7- Considering Security, Governance, and Compliance for Generative AI

Topics Covered:

Security

Protect models and data with encryption, network controls and access controls management.

IAM : Identity Access Management.

Governance : Having Policies and oversight for AI.

Compliance : Following Laws and Regulations.

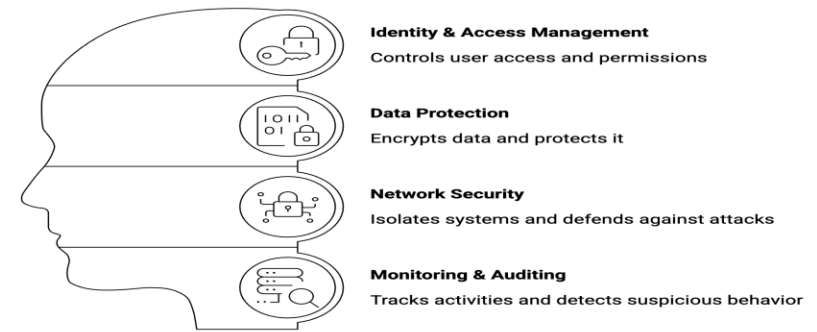
AWS Audit Manager

-> AWS Audit Manager helps organizations continuously assess compliance requirements and generate audit-ready evidence.

Skills Gained:

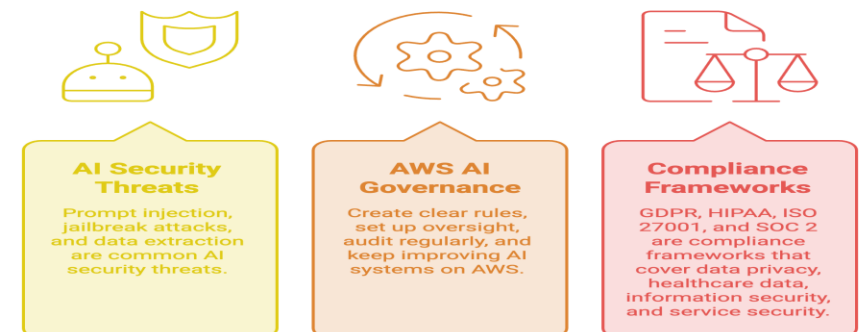
- Understanding security challenges associated with Generative AI.
- Managing access control concepts using AWS IAM.
- Understanding data protection and encryption mechanisms.
- Applying governance practices for AI systems.
- Familiarity with compliance frameworks and AWS security services.

AWS Security Layers Overview



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AI Security Threats and AWS Governance



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Sub-Course 8- Implementing Generative AI Applications

Topics Covered:

The Generative AI Applications lifecycle follows a structured approach from concept to deployment.

Problem Definition : Define the use cases and outcomes.

Model Selection : Choosing or Fine-Tuning FM's

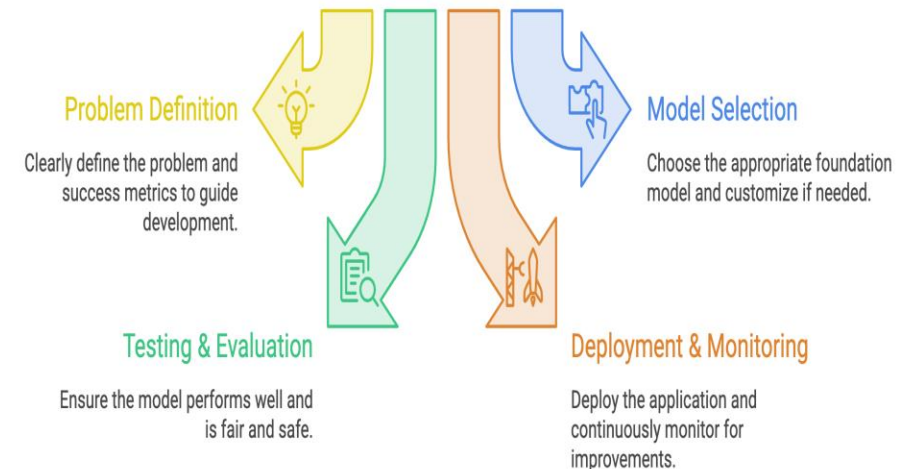
Testing and Evaluation : Test and evaluate Performance, bias and safety.

Deployment and Monitoring : Deploy and continuously monitor Applications.

Skills Gained:

- Understanding the Generative AI application lifecycle.
- Selecting AWS services based on business requirements.
- Understanding the role of AI Agents in intelligent automation.
- Evaluating AI solutions for deployment and monitoring.
- Familiarity with Amazon Bedrock, SageMaker, and Amazon Q.

How to manage the generative AI application lifecycle?



Sub-Course 9- Bridging To Certification

Topics Covered:

Importance of Certification :

-> AWS certifications validate technical expertise and enhance career opportunities in the rapidly growing AI/ML field.

Key Certifications

Focus on relevant AWS AI/ML certifications, such as the AWS Certified ML-Specialty.

Exam Preparation Strategies

Effective study techniques, practice exams and understanding the exam blueprint.

Continuous Learning

Staying updated with the latest advancements in AWS AI services and broader Gen-AI Ecosystem.

Skills Gained:

- Understanding AWS AI certification pathways.
- Familiarity with AWS certification preparation resources.
- Developing strategies for exam preparation and continuous learning.
- Understanding the importance of industry-recognized certifications.
- Building a roadmap for future professional development in AI and cloud technologies.

Comprehensive Guide to AWS AI Certification

Skills Gained
Outlines the knowledge and abilities acquired

Exam Preparation Strategies
Emphasizes effective study techniques and practice

Importance of Certification
Validates expertise and boosts career opportunities



Continuous Learning
Stresses staying updated with AI advancements

Key Certifications
Focuses on relevant AWS AI/ML certifications

Technical Knowledge and Skills Gained

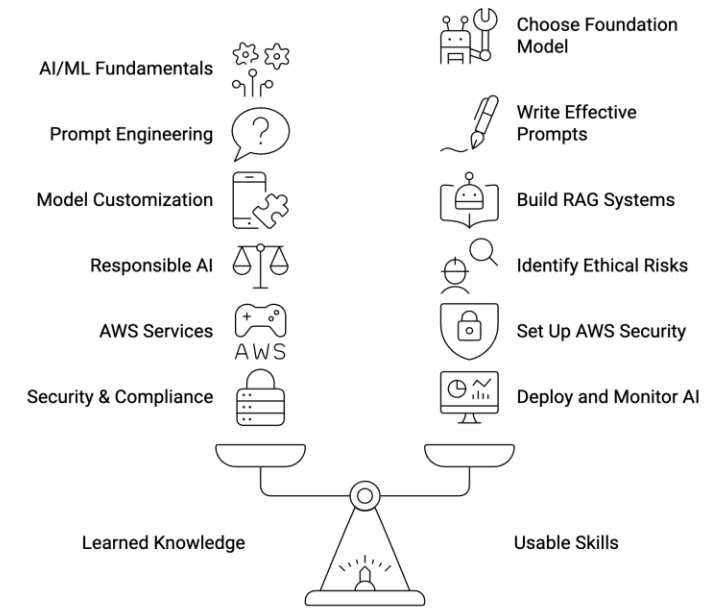
Knowledge Acquired:

- Foundational concepts of AI, ML, DL, and Generative AI.
- Principles of Foundation Models and their applications.
- Advanced Prompt Engineering techniques (Zero-shot, Few-shot, CoT).
- Retrieval Augmented Generation (RAG) architecture and implementation.
- Responsible AI principles, risks, and mitigation strategies.
- AWS services for Generative AI (Amazon Bedrock, Amazon Q, IAM).
- Security, governance, and compliance in AI applications.

Skills Developed:

- Ability to select appropriate Foundation Models for diverse tasks.
- Proficiency in designing and optimizing prompts for desired AI outputs.
- Skill in tuning inference parameters to control model behavior.
- Competence in implementing RAG systems for enhanced accuracy and context.
- Capability to identify, assess, and mitigate ethical risks in AI systems.
- Expertise in configuring AWS security and access management for AI workloads.

Balancing AI Knowledge and Skills



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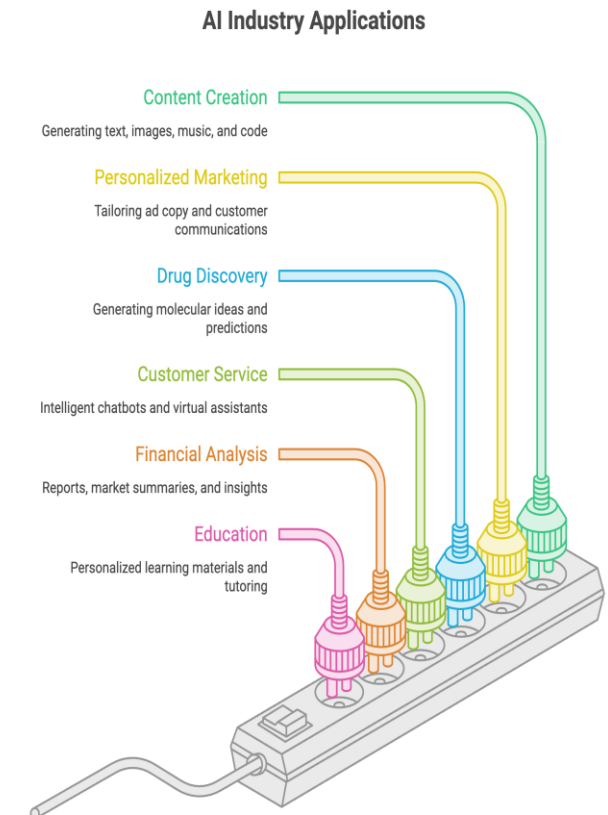
Real World and Real time Applications of the Course

Industry Use Cases :

- **Content Creation:** Generating text, images, music, and code.
- **Personalized Marketing:** Tailored ad copy and customer communications.
- **Drug Discovery & Healthcare:** Generating molecular ideas and predictions.
- **Customer Service:** Intelligent chatbots and virtual assistants.
- **Financial Analysis:** Reports, market summaries, and insights.
- **Education:** Personalized learning materials and tutoring.

Impact :

- **Enhanced Creativity:** Tools for rapid prototyping and innovation.
- **Increased Efficiency:** Automating repetitive tasks for strategic focus.
- **Personalized Experiences:** Highly customized content and services.
- **Accelerated Innovation:** Driving breakthroughs in research and tech.



Conclusion

Key Takeaways from the Specialization :

- **AI is Everywhere:** From chatbots to drug discovery, AI is changing every industry
- **AWS Makes It Possible:** Services like Bedrock make building AI applications accessible
- **Prompting is a Skill:** Learning to communicate with AI is as important as the technology itself
- **Responsibility Matters:** Building fair and safe AI is our responsibility
- **Security is Essential:** Protecting AI systems is critical
- **Certification Validates Learning:** AWS certification proves your skills

Why This Course Matters :

- Provides a strong foundation in AI and Generative AI concepts.
- Introduces AWS tools used for building intelligent applications.
- Develops industry-relevant cloud and AI skills.
- Promotes responsible and secure AI practices.
- Prepares learners for future AI careers and certifications.

Future Potential :

- Growing adoption of Generative AI across industries.
- Advancement of AI Agents and Foundation Models.
- Expansion of intelligent and automated applications.
- New opportunities in AI development and innovation.



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Course Completion Certificate



Virat P K Gupta

Certificate of Completion for
AWS Academy Graduate - Generative AI Foundations - Training Badge

Course hours completed
12 Hours

Issued on
May 20, 2026

Digital badge
<https://www.credly.com/badges/519f0892-404a-4131-b16b-6fb110c58d47>





THANK YOU